

Mathematical Foundations of Machine Learning Cheat Sheet

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Matrices and Vectors

Vector Addition: $x = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ $y = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ $x + y = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

Scalar Multiplication: if $x = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ then $2x = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$

Dot Product: $\hat{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ $\hat{w} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ $v \cdot w = 3 \times 4 + 4 \times 3 = 24$

- Distributive $u^T(v + w) = u^T v + u^T w$
- Commutative: $u^T v = v^T u$
- Not Associative: $u^T(v^T w) \neq (u^T v)^T w$
- Vector Length/Magnitude: $\|v\| = \sqrt{3^2 + 4^2} = 5$

Unit Vectors: $\mu = \frac{1}{\|v\|}$

Matrix Addition: $A + B = \begin{bmatrix} a_1 1 + b_1 1 & a_1 2 + b_1 2 \\ a_2 1 + b_2 1 & a_2 2 + b_2 2 \end{bmatrix}$

Matrix Scalar Multiplication $\delta A = \begin{bmatrix} \delta a_{11} & \delta a_{12} \\ \delta a_{21} & \delta a_{21} \end{bmatrix}$

Transpose: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

Matrix Trace: $A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 5 & 6 \\ 4 & 6 & 8 \end{bmatrix}$ $\text{trace}(A) = 16$

Matrix Multiplication:

$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 2 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, $AB = \begin{bmatrix} 2 & 7 \\ 2 & 9 \end{bmatrix}$

Properties of Determinants

- $\det(I) = 1$
- $\det(A) = 0$ when two rows are equal
- $\det(A) = 0$ when matrix has row of 0's
- if A is singular $\det(A) = 0$, if A is invertible $\det(A) \neq 0$
- $\det(AB) = \det(A)\det(B)$
- $\det(A^T) = \det(A)$

The Determinant

Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

$\det(A) = (-1)^{1+1} \cdot 1 \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} + (-1)^{1+2} \cdot 2 \begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix} + (-1)^{1+3} \cdot 3 \begin{vmatrix} 3 & 1 \\ 0 & 0 \end{vmatrix} = 1(1-0) - 2(3-0) + 3(0-0) = -5$

The Inverse Matrix

Invertibility Test

- Must have non-zero pivots after elimination
- the $\det(A)$ must be non-zero
- $Ax = 0$, where $x = 0$ must be the only solution.

Gauss-Jordan Elimination $[A|I] \Rightarrow [I|A]$

e.g. for $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$

$= \frac{1}{\det(A)} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$

e.g. for $A = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 4 & 5 \\ 6 & 7 & 7 \end{bmatrix}$ Minors = $\begin{bmatrix} -7 & -2 & +4 \\ +7 & +1 & -5 \\ +6 & +1 & -4 \end{bmatrix}$

Cofactors = $\begin{bmatrix} -7 & +2 & +4 \\ -7 & +1 & +5 \\ +6 & -1 & -4 \end{bmatrix}$

Adjugate = $Cofactors^T = \begin{bmatrix} -7 & -7 & +6 \\ +2 & +1 & -1 \\ +4 & +5 & -4 \end{bmatrix}$

$A^{-1} = \begin{bmatrix} -7 & -7 & 6 \\ 2 & 1 & -1 \\ 4 & 5 & -4 \end{bmatrix}$

Spaces/Subspaces

The four fundamental subspaces of $A_{m \times n}$

Column Space $C(A) \in \mathbb{R}^m$ $Ax = b$
 Null Space $N(A) \in \mathbb{R}^n$ $Ax = 0$
 Row Space $C(A^T) \in \mathbb{R}^m$ $A^T x = b$
 Left Null Space $N(A^T) \in \mathbb{R}^n$ $A^T x = 0$

The complete solution to $Ax = b$

- set all free variables to 0, solve $Ax=b$ for pivot variables to find $X_{particular}$.
- find the null space: $X_{nullspace}$
- the complete solution is $x = x_{particular} + x_{nullspace}$

To solve for matrix rank, Gaussian Elimination until REF, count the pivots. Linear Independence: z cannot be expressed as a linear combination of x and y .

Projections

The projection of b onto a line

$p = \frac{a^T b}{a^T a} a$ $P = \frac{a a^T}{a^T a}$

The projection of b onto the subspace spanned by the columns of A

$p = A(A^T A)^{-1} A^T b$ $P = A(A^T A)^{-1} A^T$

Two vectors are orthogonal when their dot product is 0.

If A has linearly independent columns then $A^T A$ is invertible

Gram-Schmidt Process

- Let $A=a$
- $B = b - \frac{A^T b}{A^T A} A$
- $C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B$
- $q_1 = \frac{A}{\|A\|}$ $q_2 = \frac{B}{\|B\|}$ $q_3 = \frac{C}{\|C\|}$

Matrix Decomposition

Finding Eigenvalues $\det(A - \lambda I) = 0$

Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ solve for $\det(\begin{bmatrix} 1 - \lambda & 0 \\ 0 & 2 - \lambda \end{bmatrix})$

$$(1 - \lambda)(2 - \lambda) = 0$$

$$\lambda = 1, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = 0, v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda = 2, \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -v_1 \\ 0 \end{bmatrix} = 0, v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Compute the SVD

A is given, $A = U\Sigma V^T$

$$\det(A^T A - \lambda I) = 0$$

V is the matrix composed of v_1 and v_2 from calculating the determinant

$$\Sigma = \begin{bmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \end{bmatrix}$$

U is found by calculating $u_1 = \frac{Av_1}{\delta_1}$, $\delta_1 = \frac{1}{\sqrt{\lambda_1}}$ and

$$u_2 = \frac{Av_2}{\delta_2}, \delta_2 = \frac{1}{\sqrt{\lambda_2}}$$